

Order Management System

Build a RESTful Order Management System where users can manage customers, products, and orders. The system should follow a multi-layer architecture and use best practices such as Spring Boot, JPA, and exception handling.

1. Customer Management:

- Add, update and fetch customers.
- A customer should have the following fields:
 - `id` (auto-generated)
 - `name` (String, required)
 - `email` (String, unique, required)
 - `phone` (String, unique, required)

2. Product Management:

- Add, update and fetch products.
- A product should have the following fields:
 - `id` (auto-generated)
 - `name` (String, required)
 - `price` (BigDecimal, required)
 - `stock` (Integer, required, greater than or equal to 0)

3. Order Management:

- Place a new order. An order should:
 - Contain customer details.
 - Contain one or more products with quantities.
 - Validate if sufficient stock is available before placing the order.
 - Deduct stock upon successful order placement.
- Fetch orders by customer ID.
- Calculate the total order amount.

4. Reporting:

- Provide an API to fetch:
 - Total orders placed by each customer.
 - Top 5 customers based on the number of orders.